

CS5239 Computer System Performance Analysis

Lstats: Review of Probability Distributions

Outline

- Discrete and Continuous Random Variables
- Expectation
 - Expected Value (Population Mean), Variance and Standard Deviation
- Statistical Distributions: Discrete and Continuous
- Poisson Arrival and Exponential Inter-arrival
- Summary
- Key Terms

Objectives

- use of probability distributions
 - model workload, service centers
 - solve queuing models
- review of statistical models and common probability distributions

L06: Queuing Models

A Typical Queuing Problem

service
intensity

service
demand

A storage system consists of three disk drives sharing a common queue. The **average time to serve an I/O request is 50 milliseconds**. The I/O requests arrive to the storage system at the rate of **30 requests per second**. Use a **M/M/3** model for the system, determine the average disk drive utilization, the probability of queuing and the average number of jobs in the system.

exponential inter-
arrival time and
exponential service
time

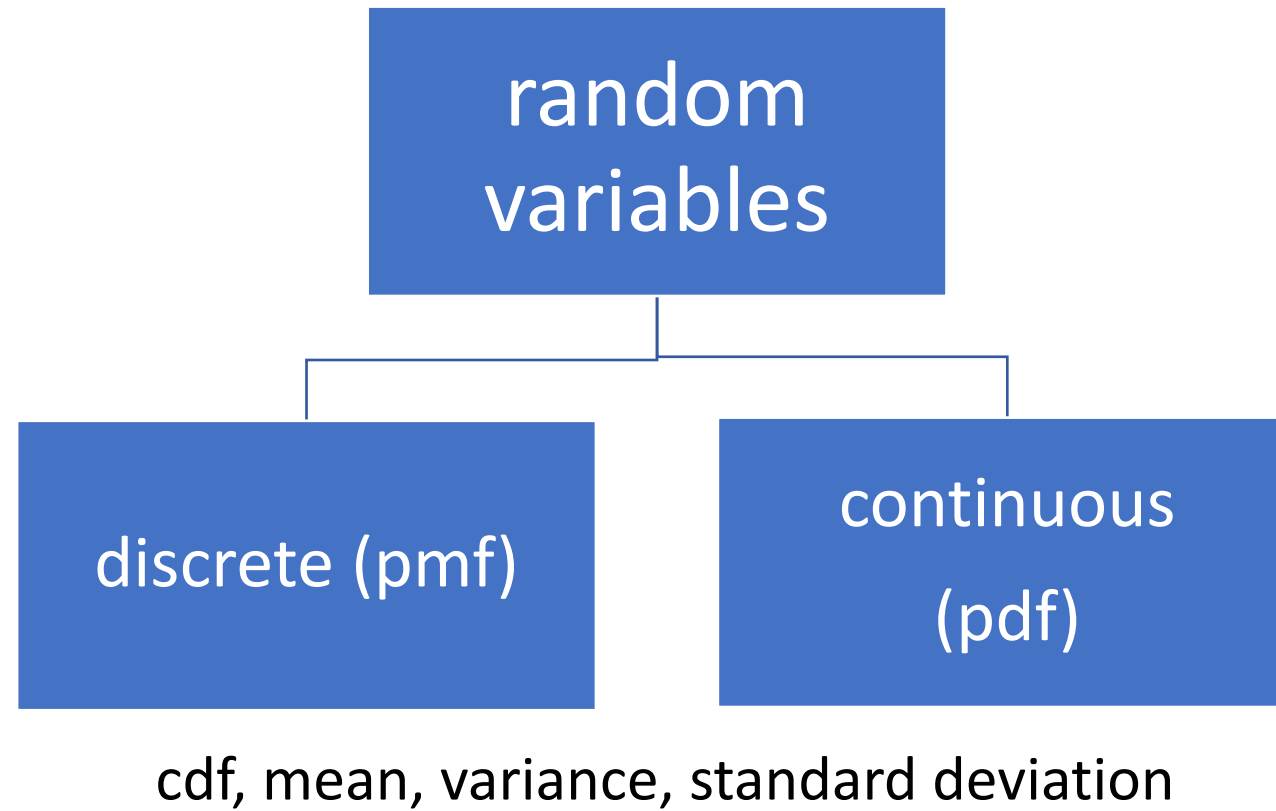
Statistical Models

- performance of a system depends on its *workload*
- workload (model) required as input to capacity planning, analytic models and simulation
- probability concepts required to *model* and *characterize* the workload, and for the analytic models
- **experiment**: a process whose outcome is not known with certainty
- **sample space**: set of all possible outcomes of an experiment
- **random variable**
 - a real-valued function of the outcome of an experiment
 - a function whose domain is the **sample space** and whose range is the set of numbers (or simply some numerical outcome of our experiment)

Example: Toss a die

Sample space = {1, 2, 3, 4, 5, 6}

Random variable $X = 1, X = 2, \dots$



Review of Terminology and Concepts

- Discrete random variable (pmf)
- Continuous random variable (pdf)
- Cumulative distribution function (cdf)
- Expectation

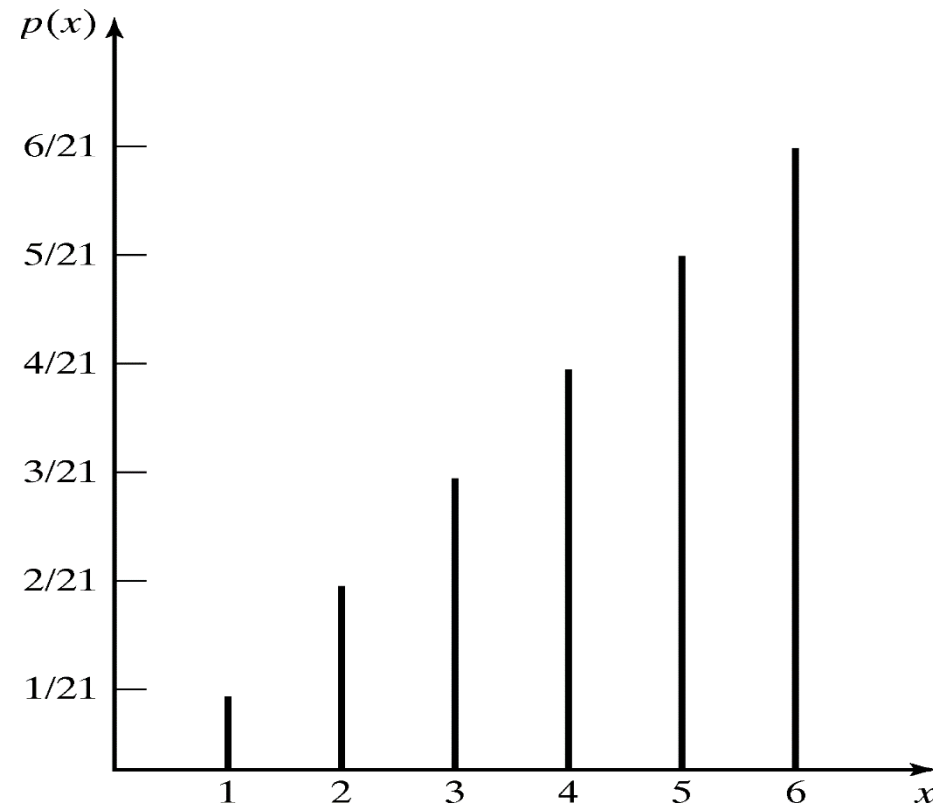
Discrete vs Continuous Random Variable

- a ***discrete*** random variable can take at most *countably* infinite number of possible values vs a ***continuous*** random variable can take an *uncountable* set of possible values
- Which of these random variables are discrete or continuous?
 - sum of rolls of two dice
 - number of http request arrivals at a web site at time t
 - time until the next arrival at a website
 - CPU time required to process a HTTP request

Discrete Random Variable

- X is a discrete random variable if the number of possible values of X is **finite**, or *countably infinite*
- Example: Consider jobs arriving at a computer system
 - Let X be the number of jobs arriving each week at a computer system
 - R_x = possible values of X (range space of X) = $\{0, 1, 2, \dots\}$
 - $p(x_i)$ = probability the random variable is $x_i = P(X = x_i)$
 - $p(x_i)$, $i = 1, 2, \dots$ must satisfy:
 1. $p(x_i) \geq 0$, for all i
 2. $\sum_{i=1}^{\infty} p(x_i) = 1$
 - Collection of pairs $[x_i, p(x_i)]$, $i = 1, 2, \dots$, is called the probability distribution of X , and $p(x_i)$ is called the **probability mass function** (pmf) of X

Example: Discrete Random Variable



Probability Mass Function for a Die

(Probability that a given face lands up is proportional to the number of spots showing)

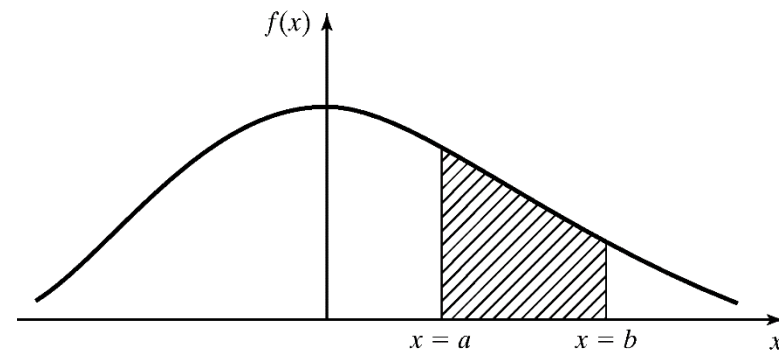
Continuous Random Variable

- X is a continuous random variable if X takes on an *uncountable set of possible values*
- Probability that X lies in the interval $[a, b]$ is:

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

- $f(x)$ denotes the **probability density function** (pdf) of X , satisfies:

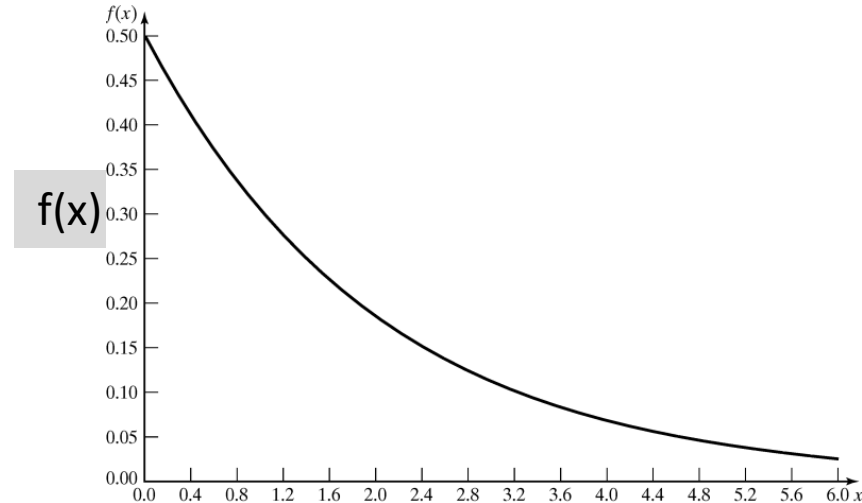
1. $f(x) \geq 0$, for all x in R_X
2. $\int_{R_X} f(x)dx = 1$
3. $f(x) = 0$, if x is not in R_X



Example: Continuous Random Variable

If the life of a device is a *continuous* random variable, X , with the following pdf:

$$f(x) = \begin{cases} \frac{1}{2}e^{-x/2}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$



- X has an exponential distribution with mean 2 years
- Probability that the device's life is between 2 and 3 years is:

$$P(2 \leq x \leq 3) = \frac{1}{2} \int_2^3 e^{-x/2} dx = 0.14$$

CUMULATIVE DISTRIBUTION FUNCTION

- $F(x)$ denotes the Cumulative Distribution Function (cdf) where $F(x) = P(X \leq x)$

- if X is **discrete**

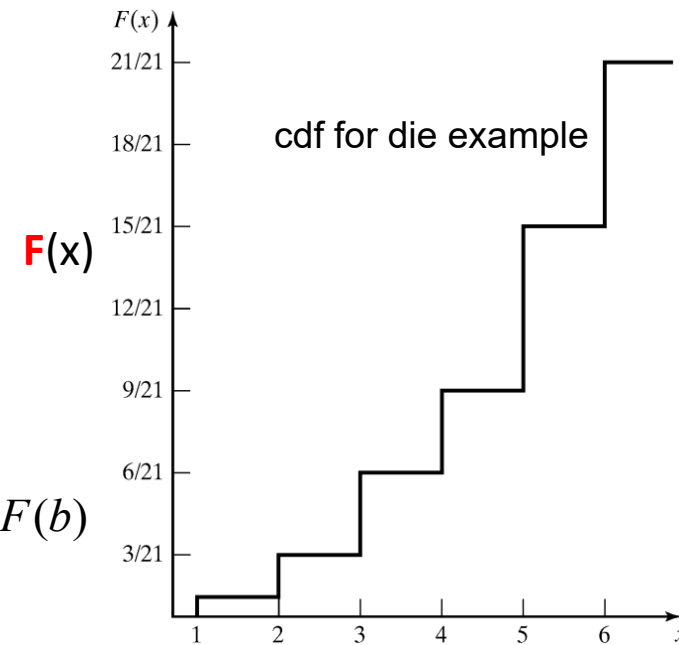
$$F(x) = \sum_{\text{all } x_i \leq x} p(x_i)$$

- if X is **continuous**

- Properties

$$F(x) = \int_{-\infty}^x f(t)dt$$

1. F is nondecreasing function. If $a < b$, then $F(a) \leq F(b)$
2. $\lim_{x \rightarrow \infty} F(x) = 1$
3. $\lim_{x \rightarrow -\infty} F(x) = 0$



- Probability question about X can be answered using the cdf, example:

$$P(a < X \leq b) = F(b) - F(a), \text{ for all } a < b$$

Example: Cumulative Distribution Function

The lifetime of a device has the following cdf:

$$F(x) = \frac{1}{2} \int_0^x e^{-t/2} dt = 1 - e^{-x/2}$$

- Probability that the device lasts for less than 2 years:

$$P(0 \leq X \leq 2) = F(2) - F(0) = F(2) = 1 - e^{-1} = 0.632$$

- Probability that it lasts between 2 and 3 years:

$$P(2 \leq X \leq 3) = F(3) - F(2) = (1 - e^{-(3/2)}) - (1 - e^{-1}) = 0.145$$

Expectation

- mean of a distribution is a.k.a. expectation and for a random variable X , $E(X)$ denotes the mean

- If X is discrete,

$$E(X) = \sum_{\text{all } i} x_i p(x_i)$$

- If X is continuous,

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

- Variance of X is denoted as $V(X)$, $\text{var}(X)$ or σ^2
 - $V(X) = E[(X - E[X])^2]$ or $V(X) = E(X^2) - [E(x)]^2$
 - measures the spread (variation) of the possible values of X around the mean, i.e., bet
- Standard deviation of X is denoted by σ
 - Definition: square root of $V(X)$
 - Expressed in the same units as the mean

Example: Expectation

Mean of life of the device is:

$$E(X) = \frac{1}{2} \int_0^{\infty} x e^{-x/2} dx = -x e^{-x/2} \Big|_0^{\infty} + \int_0^{\infty} e^{-x/2} dx = 2$$

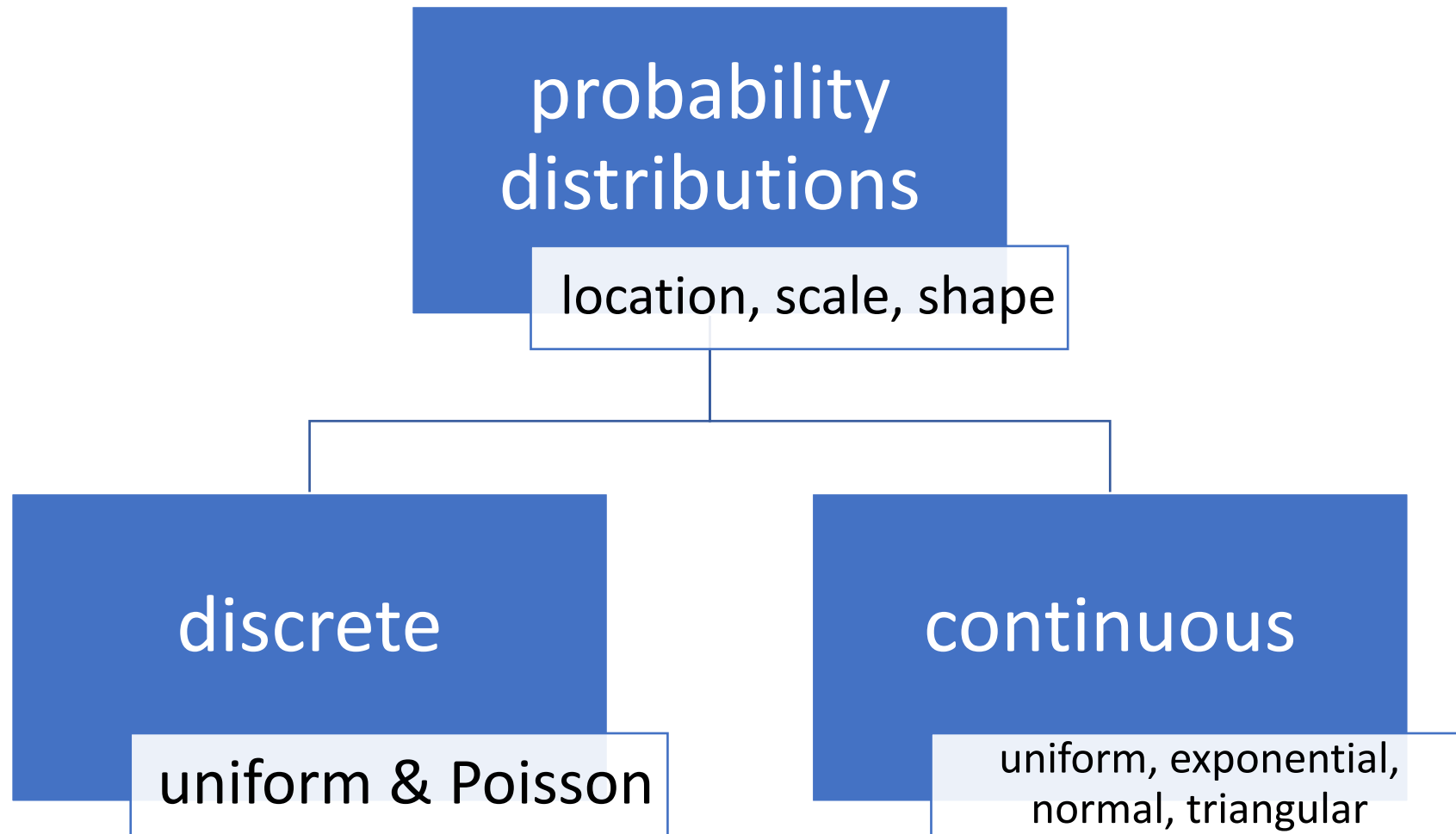
To compute variance of X , we first compute $E(X^2)$:

$$E(X^2) = \frac{1}{2} \int_0^{\infty} x^2 e^{-x/2} dx = -x^2 e^{-x/2} \Big|_0^{\infty} + \int_0^{\infty} x e^{-x/2} dx = 8$$

Hence, the variance and standard deviation of the device's life are:

$$V(X) = 8 - 2^2 = 4$$

$$\sigma = \sqrt{V(X)} = 2$$



REVIEW OF STATISTICAL DISTRIBUTIONS

- Common (parametric) distributions – random variable follows a particular (known) distribution
- Parameters:
 - **Location:** specifies an abscissa (x-axis) location point of a distribution's range of values. As location changes, the associated distribution shifts left or right.
 - **Scale:** determines scale (or unit) of measurement of the values in the range of the distribution. A change in scale compresses or expands the associated distribution without altering its basic form.
 - **Shape:** determines the basic form or shape of a distribution within the general family of distributions of interest. A change in shape alters a distribution properties such as skewness.

Review of Statistical Distributions

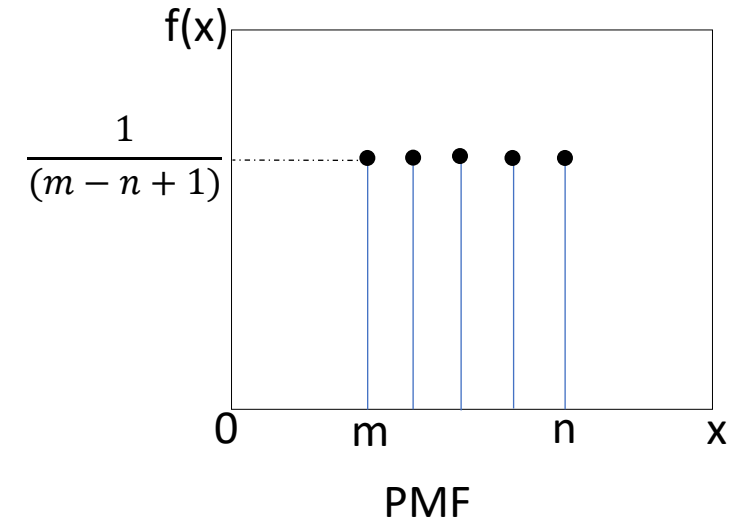
- Discrete distributions
 - Uniform(m, n) where m (location), $n-m$ (scale)
 - Poisson(λ) where λ (mean)
- Continuous distributions
 - Uniform(a, b) where a (location), $b-a$ (scale)
 - Exponential(α) where α (scale)
 - Normal(μ, σ^2) where μ (location), σ (scale)
 - Weibull (v, α, β) where v (location), α (scale), β (shape)
 - Triangular (a, b, c) where a (location), $c-a$ (scale), b (shape)
- Poisson arrival and exponential inter-arrival

Discrete Distributions

- Uniform and Poisson

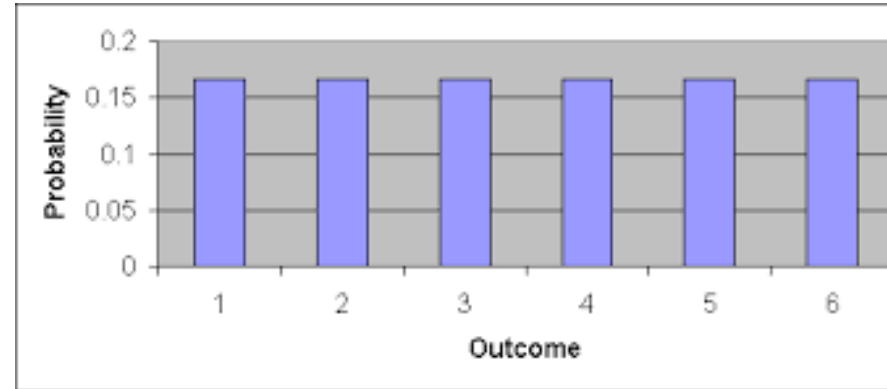
Uniform Distribution (Discrete): **UD(m, n)**

- Parameters: m = lower limit, an integer
 n = upper limit, $n > m$, an integer
- Range: $x = m, m + 1, m + 2, \dots, n$
- pmf: $f(x) = \frac{1}{(n-m+1)}$
- cdf: $F(x) = \begin{cases} 0 & \text{if } x < m \\ \frac{x-m+1}{n-m+1} & \text{if } m \leq x < n \\ 1 & \text{if } n \leq x \end{cases}$
- Mean: $\frac{n+m}{2}$
- Variance: $((n - m + 1)^2 - 1)/12$



Example: UD(m, n)

Throwing a dice:
m=1, n=6



$$Mean = \frac{n + m}{2} = \frac{7}{2} = 3.5$$

$$\begin{aligned} Variance &= ((n - m + 1)^2 - 1) / 12 \\ &= ((6 - 1 + 1)^2 - 1) / 12 = 2.92 \end{aligned}$$

Poisson Distribution: **Poisson**(λ)

- Commonly used in queuing models to model the number of arrivals over a given interval

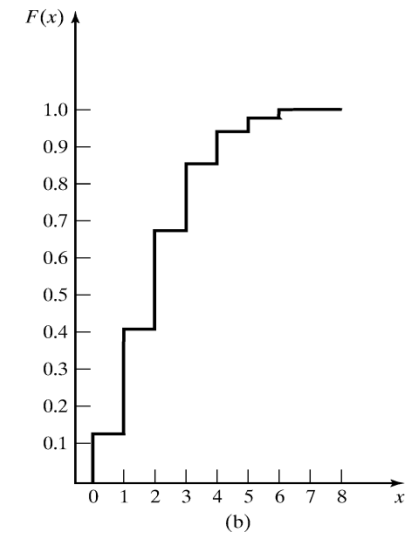
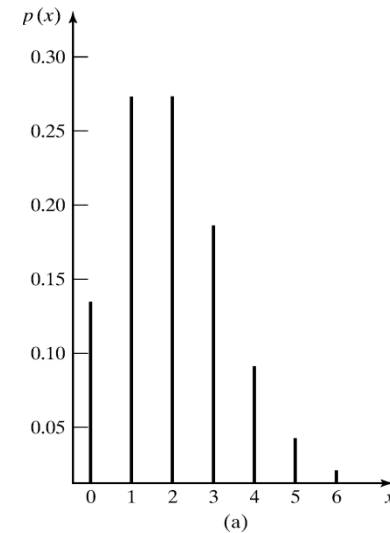
- Key characteristics

- Parameters:
 - λ is the mean number of events within a time interval, i.e., $\lambda = \text{mean}$, $\lambda > 0$

- Range: $x = 0, 1, 2, \dots, \infty$

- pmf: $f(x) = P(X = x) = \lambda^x \frac{e^{-\lambda}}{x!}$

- Mean and variance are both equal to λ



Example: Poisson(λ)

The number of visitors to a webserver per minute follows a Poisson distribution. If the average number of visitors per minute is 4, what is the probability that

- i. there are two or fewer visitors in one minute?
- ii. there are exactly two visitors in 30 seconds?

Example: Poisson(λ)

- (i) If the average number of visitors in a minute is 4, $\lambda = 4$

Calculate $P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$

$$P(X = 0) = \frac{e^{-4} 4^0}{0!} = e^{-4}$$

$$P(X = 1) = \frac{e^{-4} 4^1}{1!} = 4e^{-4}$$

$$P(X = 2) = \frac{e^{-4} 4^2}{2!} = 8e^{-4}$$

Thus, the probability of two or fewer visitors in a minute is $e^{-4} + 4e^{-4} + 8e^{-4} = 0.238$

Example: Poisson(λ)

- (ii) If the average number of visitors in 1 minute is 4 then the average number of visitors in 30 seconds is 2.

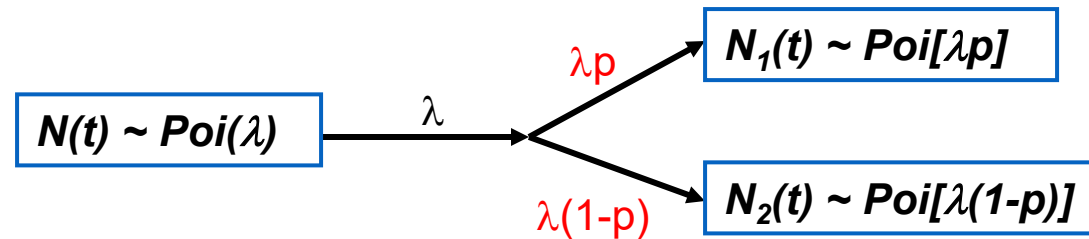
Thus, $\lambda = 2$ and
$$P(X = 2) = \frac{e^{-2}2^2}{2!} = 2e^{-2} = 0.271$$

Splitting and Pooling

Consider a Poisson process $\{N(t), t \geq 0\}$ with (arrival) rate of λ

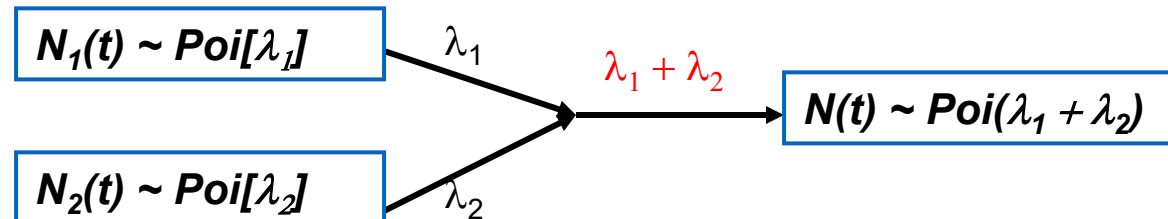
- Splitting

- Suppose each event in the Poisson process is divided as Type I, with probability p and Type II, with probability $1-p$ and $N_1(t)$ and $N_2(t)$ denote the random variables respectively
- Then, $N(t) = N_1(t) + N_2(t)$, where $N_1(t)$ and $N_2(t)$ are both Poisson processes with rates λp and $\lambda(1-p)$



- Pooling

- Similarly, suppose two Poisson processes are pooled together
- Then, $N_1(t) + N_2(t) = N(t)$, where $N(t)$ is a Poisson processes with rates $\lambda_1 + \lambda_2$



Continuous Distributions

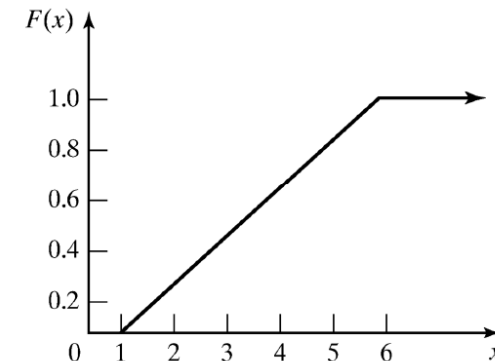
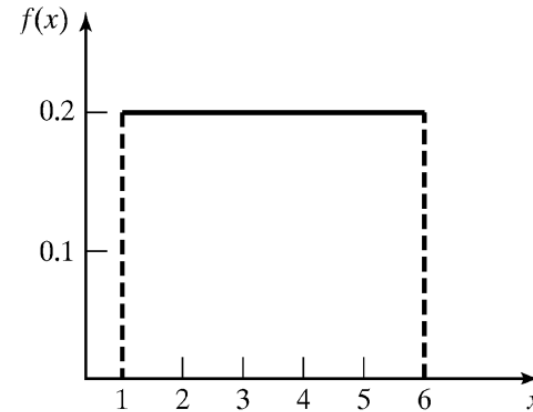
- Uniform, Exponential, Normal & Triangular

Uniform Distribution (Continuous): **U(a, b)**

- Commonly used when a random variable is bounded and no further information is available, e.g. seek time on a disk

- Key characteristics

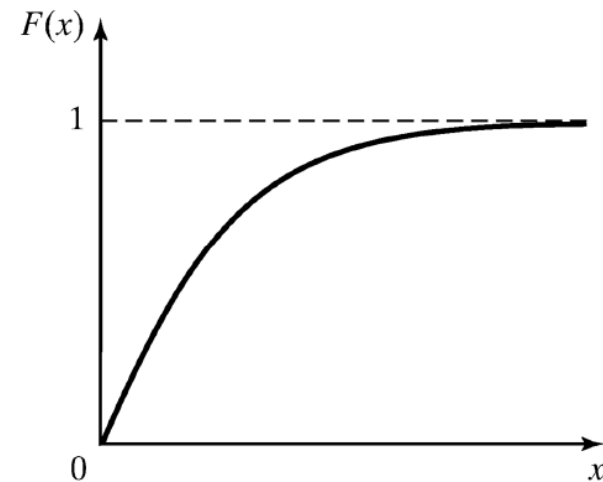
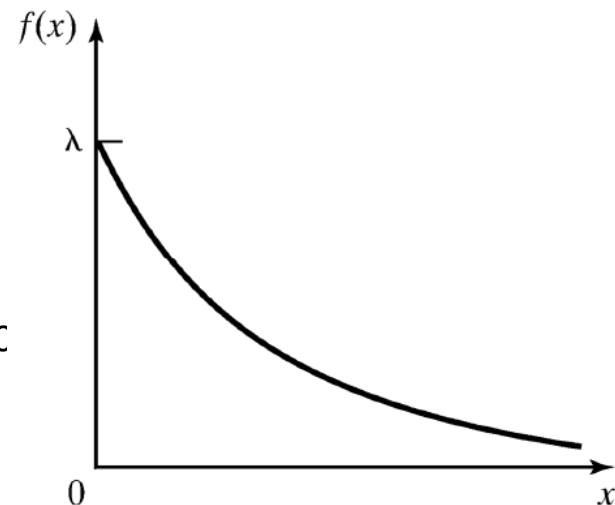
- Parameters: a = lower limit, b = upper limit
- Range: $a \leq x \leq b$
- PDF: $f(x) = \frac{1}{(b-a)}$
- CDF: $F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x < b \\ 1 & \text{if } b \leq x \end{cases}$
- Mean: $\frac{a+b}{2}$
- Variance: $(b-a)^2/12$



- U(0, 1): generates random numbers to obtain random variates

Exponential Distribution: **exp**(α)

- Extensively used in queueing models, e.g., α models the inter-arrival time of jobs
- Key characteristics
 - Parameters: α = scale parameter = mean ($= \frac{1}{\lambda}$ and > 0)
 - If inter-arrival times of jobs are exponentially distributed with mean of $\frac{1}{\lambda}$ (mean time to the next arrival)
 - pdf: $f(x) = \lambda e^{-x\lambda}$
 - cdf: $F(x) = 1 - e^{-x\lambda}$
 - range: $0 \leq x \leq \infty$
 - Variance: $\frac{1}{\lambda^2}$
 - Memoryless



Example: Exponential Distribution

Suppose the life of a device is $\text{exp}(\lambda = 1/3 \text{ per hour})$, hence, on average, 1 failure per 3 hours

a. probability that the lamp lasts longer than its mean life is: $P(X > 3) = 1 - (1 - e^{-3/3}) = e^{-1} = 0.368$

b. probability that the lamp lasts between 2 to 3 hours is:

$$P(2 \leq X \leq 3) = F(3) - F(2) = 0.145$$

c. memoryless property: **$P(X > s+t \mid X > s) = P(X > t)$**

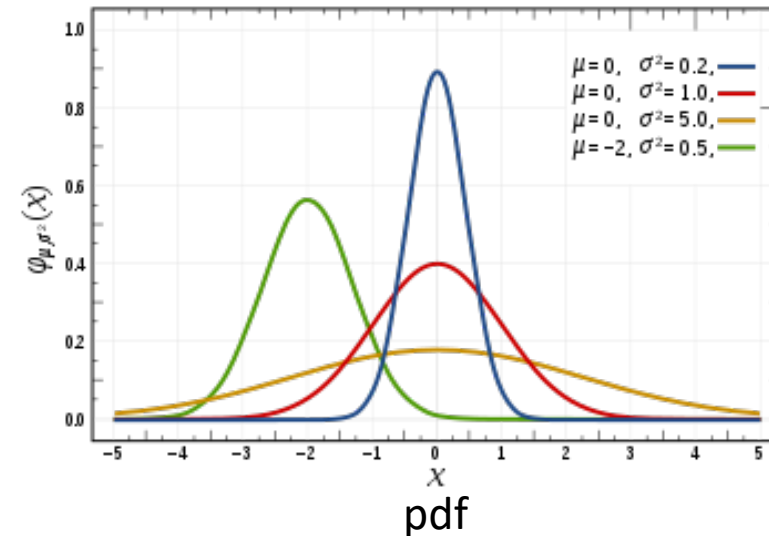
Example: probability that it lasts for another hour given it is operating for 2.5 hours:

$$P(X > 3.5 \mid X > 2.5) = P(X > 1) = e^{-1/3} = 0.717$$

Normal Distribution $\mathbf{N}(\mu, \sigma)$

Key characteristics

- Parameters: μ = Mean,
 σ = standard deviation $\sigma > 0$
- Range: $-\infty \leq x \leq \infty$
- pdf: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$
- Mean: μ
- Variance: σ^2
- $N(0, 1)$ – unit (standard) normal distribution



Usage

- Errors in measurement
- Error in modeling to account for a number of factors that are not included in the model
- Sample means of a large number of independent observations from a given distribution

Example 1: Normal Distribution

Suppose the length of time between charges of the battery is normally distributed with a mean of 50 hours and a standard deviation of 15 hours. What is the probability that the length of time between battery charging is between 50 and 70 hours?

Solution:

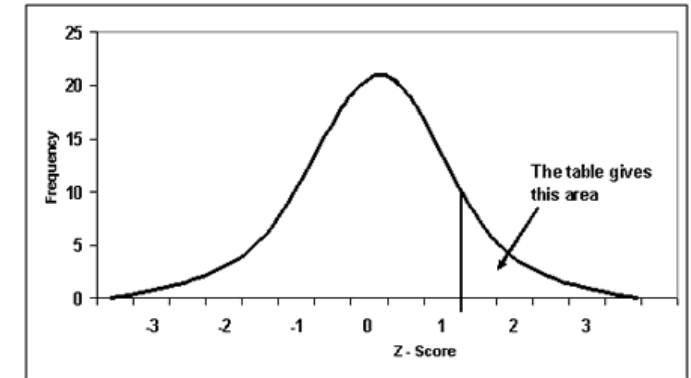
Let x be the random variable that represents the length of time. It has a mean of 50 and a standard deviation of 15. We have to find the probability that x is between 50 and 70 or $P(50 < x < 70)$

Example 1: Normal Distribution

Solution cont.:

We convert the given values to z-scores using the formula

$$z = \frac{x - \mu}{\sigma}$$



For $x = 50$, $z = (50 - 50) / 15 = 0$

For $x = 70$, $z = (70 - 50) / 15 = 1.33$ (rounded to 2 decimal places)

$P(50 < x < 70) = P(0 < z < 1.33) = [\text{area to the left of } z = 1.33] - [\text{area to the left of } z = 0]$
 $= 0.9082 - 0.5 = 0.4082$ (values are read from the z-value table[1])

The probability that John's computer has a length of time between 50 and 70 hours is equal to 0.4082.

[1] <http://www.z-table.com/>

More examples: http://www.anlyzemath.com/statistics/normal_distribution.html

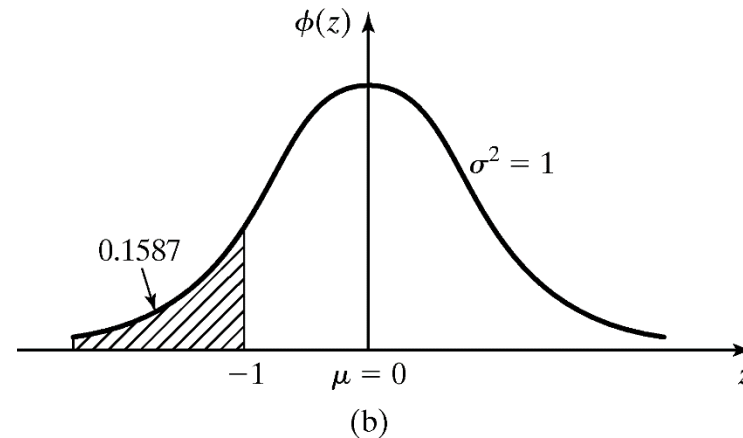
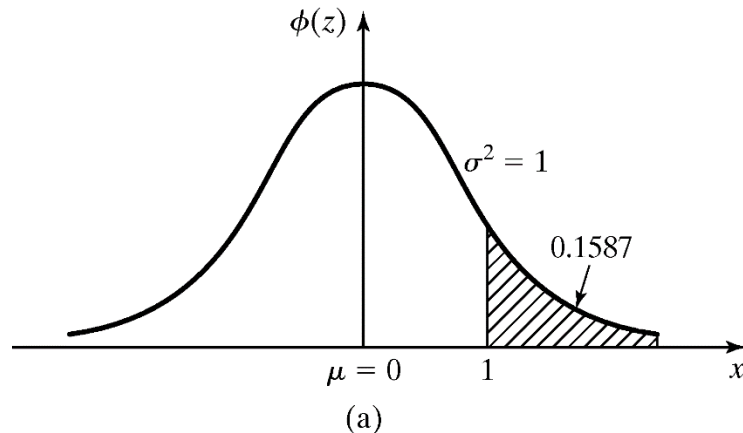
Example 2: Normal Distribution

The time required to load an ocean going vessel, X , is distributed as $N(12,4)$

The probability that the vessel is loaded in less than 10 hours:

$$F(10) = \Phi\left(\frac{10-12}{2}\right) = \Phi(-1) = 0.1587$$

Using the symmetry property, $\Phi(1)$ is the complement of $\Phi(-1)$



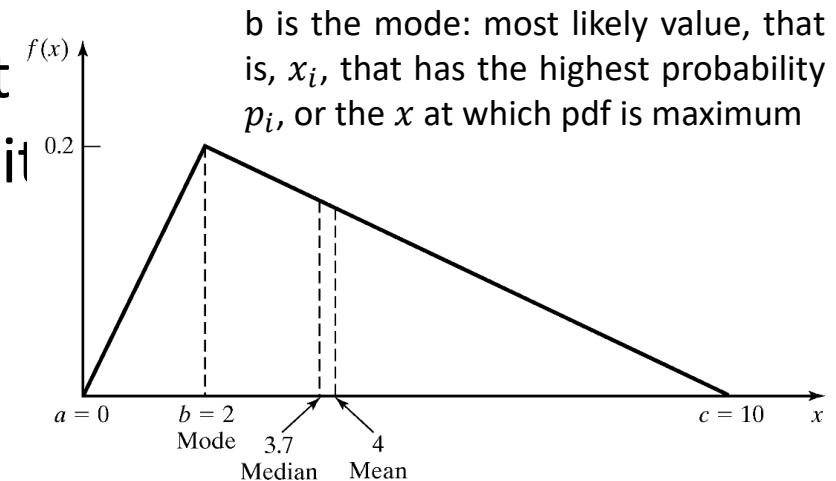
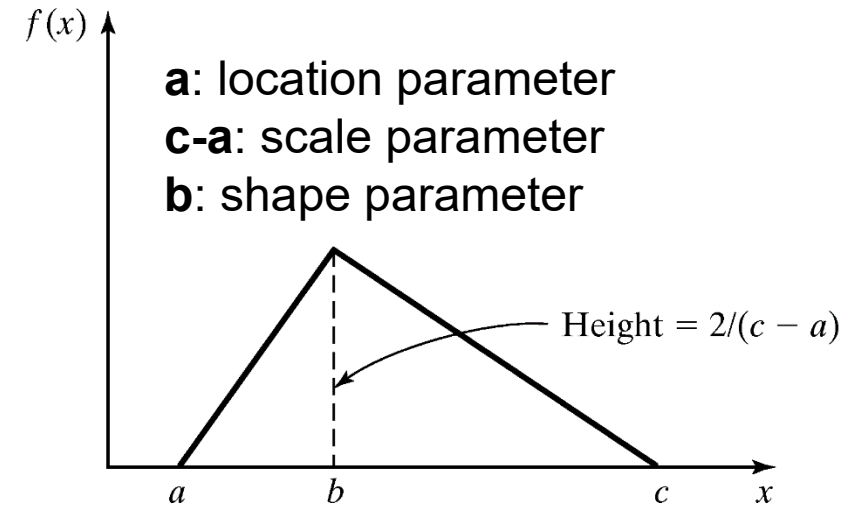
Triangular Distribution (a, b, c)

- pdf

$$f(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \leq x \leq b \\ \frac{2(c-x)}{(c-b)(c-a)} & b < x \leq c \\ 0 & \text{otherwise} \end{cases}$$

- Mean $E(X) = (a+b+c)/3$
- Variance $V(X) = (a^2+b^2+c^2-ab-ac-bc)/18$

- models a process where data is not easily obtained but knowledge of its characteristics can establish its bounds and most likely value



POISSON ARRIVAL AND EXPONENTIAL INTER-ARRIVAL

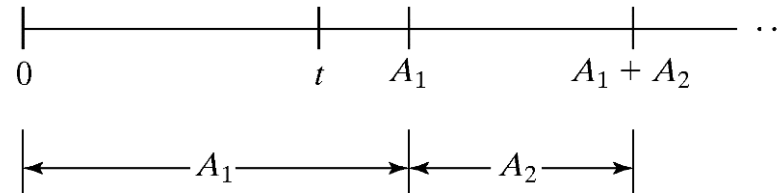
If the number of arrivals is **Poisson** distributed then the time between successive arrivals (inter-arrivals) is **exponentially** distributed with mean of $1/\lambda$ (converse is also true).

Poisson Arrival and Exponential Inter-arrival

- Poisson arrival: $N(t)$ is a counting function that represents the number of arrivals in $[0,t]$

$$P[N(t) = n] = \frac{e^{-\lambda t} (\lambda t)^n}{n!}, \quad \text{for } t \geq 0 \text{ and } n = 0, 1, 2, \dots$$

- Consider the inter-arrival times of a Poisson process $(A_1, A_2, \dots)$, where A_i is the elapsed time between arrival i and arrival $i+1$



Since the 1st arrival occurs after time t only if there are no arrivals in the interval $[0,t]$, then $\{A_1 > t\} = \{N(t) = 0\}$ and therefore, from Poisson distribution, $P\{A_1 > t\} = P\{N(t) = 0\} = e^{-\lambda t}$

Thus, the probability that the 1st arrival will occur in $[0,t]$ is

$$P\{A_1 \leq t\} = 1 - e^{-\lambda t}$$

which is the cdf of the exponential distribution with parameter λ . Hence, A_1 is exponentially distributed with mean $E(A_1) = 1/\lambda$

Summary

- Random variables: discrete versus continuous
- Probability mass (or density) function
- Cumulative density function
- Mean, variance and standard deviation
- Probability distribution: discrete and continuous
- Poisson arrival & exponential inter-arrival

References

1. The Art of Computer Systems Performance Analysis, Raj Jain, Wiley Professional Computing
2. Measuring Computer Performance - A Practitioner's Guide, David J. Lilja, Cambridge University Press
3. Discrete-Event Simulation – A First Course, Pearson

Key Terms

- Random variables
- Discrete versus continuous
- Probability mass function
- Probability density function
- Cumulative density function
- Mean, variance and standard deviation
- Discrete probability distribution
- Continuous probability distribution
- Poisson – splitting and pooling
- Poisson arrival & exponential inter-arrival

Summarizing Data by a Single Number

- **indices of central tendencies:** mean, median, mode
- **sample mean:** sum of all observations divided by the number of observations
- **sample median:** sorts observations in an increasing order and taking the observation that is in the middle of the series. If the number of observations is even, the mean of the middle two values is used as a median.
- **sample mode:** plots a histogram and takes the midpoint of the bucket where the histogram peaks. For categorical variables, mode is given by the category that occurs most frequently.
- mean and median always exist and are unique. Mode, on the other hand, may not exist.

Types of Means

- three types of means
 - Arithmetic mean
 - Harmonic Mean
 - Geometric mean
- in the earlier slides, we compute the mean for a probability distribution. Next, we look at calculating mean for a given set of (measurement) values
- selecting the most suitable type of mean should be done based on the application and context

Arithmetic Mean

Arithmetic mean is defined as

$$\bar{x}_A = \frac{1}{n} \sum_{i=1}^n x_i$$

where x_i values are the individual measurements being averaged together.

Example: Given a set of execution time measurements, the arithmetic is

$$\bar{T}_A = \frac{1}{n} \sum_{i=1}^n T_i$$

Since $\bar{T}_A$ is directly proportional to the total execution time, arithmetic mean is the best type of mean to summarize execution times

Harmonic Mean

Harmonic mean is defined as

$$\bar{x}_H = \frac{n}{\sum_{i=1}^n (1/x_i)}$$

where x_i values represent the n separate values that are being averaged together

Example: If we summarize execution rates using harmonic mean,

$$\bar{M}_H = \frac{n}{\sum_{i=1}^n (1/M_i)} = \bar{x}_H = \frac{n}{\sum_{i=1}^n (\frac{T_i}{F})} = \frac{Fn}{\sum_{i=1}^n (T_i)}$$

$\bar{M}_H$ is inversely proportional to the total execution time, thus is an appropriate mean to summarize execution rates

Geometric Mean

- Geometric mean is defined as the n th root of the product of the n individual x_i values

$$\bar{x}_G = \sqrt[n]{x_1 x_2 x_3 \dots x_i \dots x_n} = \left(\prod_{i=1}^n x_i \right)^{1/n}$$

- The key advantage of geometric mean is that it maintains consistent relationships when comparing normalized value regardless of the basis system used to normalize the measurements

Confidence Interval

- A probabilistic statement about the range that the characteristics of the system would lie
- Population characteristics (parameters) hold for all possible realizations of the stochastic process describing the system behavior
- A ***confidence interval*** gives an *estimated range* of values which is likely to include an unknown population parameter
- estimated range is derived from a given set of sample data
- often, this parameter is the population mean μ , which is estimated using the sample mean $\bar{x}$

Confidence Intervals for the Mean

- A **confidence interval** gives an estimated range of values which is likely to include an unknown population parameter, the estimated range being calculated from a given set of sample data
- Often, this parameter is the population mean μ , which is estimated through the sample mean $\bar{x}$.
- Each **sample mean** is an *estimate of the population mean*
- Given **k** samples, we have **k** different estimates
- Problem:
To obtain a **single estimate** of the population mean from these k samples (estimates)

What should be the size of k ?

Confidence Intervals for the Mean

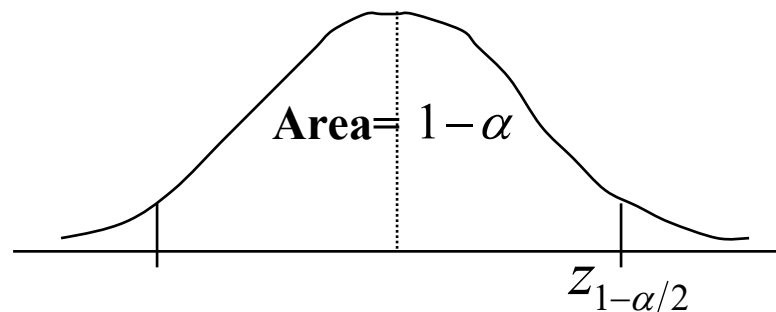
- Central Limit Theorem (CLT) $\Rightarrow$ if the n observations in a sample $\{x_1, x_2, \dots, x_n\}$ are independent and the population has a mean μ and standard deviation σ , then for large n the sample mean has a Normal distribution:

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- Standard deviation of sample mean is called standard error
- Using the CLT, we can state with $100(1 - \alpha)$ % confidence that the population mean is between $\left(\bar{x} - z_{1-\alpha/2} \frac{s}{\sqrt{n}}\right)$ and $\left(\bar{x} + z_{1-\alpha/2} \frac{s}{\sqrt{n}}\right)$
- In other words, a $100(1 - \alpha)\%$ CI for the population mean is given by

$$\left(\bar{x} - z_{1-\alpha/2} \frac{s}{\sqrt{n}}, \bar{x} + z_{1-\alpha/2} \frac{s}{\sqrt{n}}\right)$$

where $\bar{x}$ - sample mean, s - sample deviation, n - sample size, and $z_{1-\alpha/2}$ is the quantile of a unit normal variate (available in tables)



Confidence Intervals for the Mean

- Standard error $\downarrow$ as $n \uparrow$
- If the sample size is large (say, $n < 30$), CI can be constructed if the observations come from a population with Normal distribution
- The procedure is the same as for small samples except that we use the ***t-distribution*** in place of standard normal
- Thus, the $100(1 - \alpha)\%$ confidence interval is given by

$$(\bar{x} - t_{[1-\alpha/2; n-1]} s / \sqrt{n}, \bar{x} + t_{[1-\alpha/2; n-1]} s / \sqrt{n})$$

- Here, $t_{[1-\alpha/2; n-1]}$ is the $(1 - \alpha/2)$ -quantile of the t -variate with $n-1$ degrees of freedom (these quantiles are also available in tables)

Confidence Intervals for the Mean

Obtain two probability bounds c_1 and c_2 , such that there is high probability $(1 - \alpha)$ that the population mean is in the interval (c_1, c_2) :

$$P\{c_1 \leq \mu \leq c_2\} = 1 - \alpha$$

where

- (c_1, c_2) - confidence interval for the population mean
- $100(1 - \alpha)$ - confidence level (%)
- $(1 - \alpha)$ - confidence coefficient
- α - significance level, i.e. $P(\text{reject } H_0 | H_0 \text{ is true})$

Mode, Median and Quantiles

- Mode

The most likely value, that is, x_i , that has the highest probability p_i , or the x at which pdf is maximum is called the mode of x

- Median

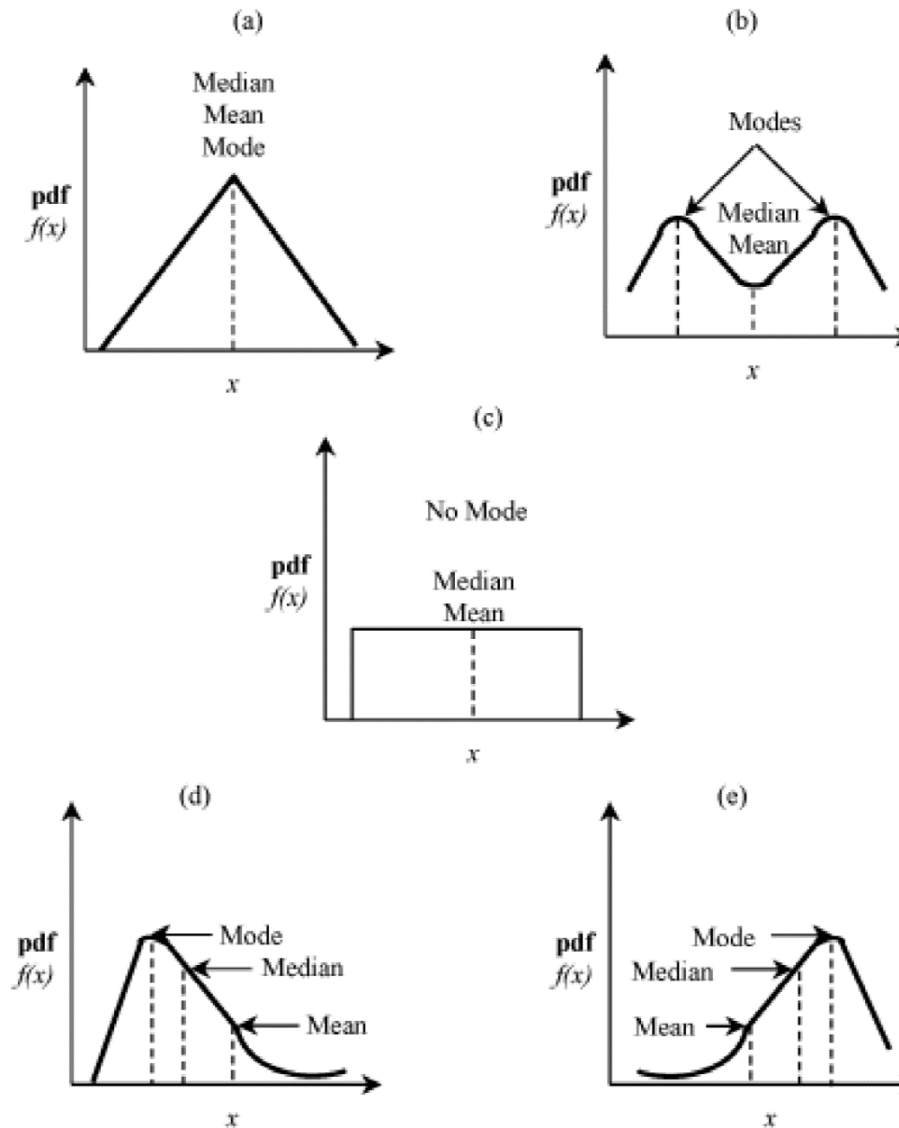
The 50-percentile (or 0.5 quantile) of a random variable is called its median

- Quantile

The x value at which the cdf takes a value α is called the α -quantile or 100α -percentile. It is denoted by x_α and is such that the probability of x being less than or equal to x_α is α

$$P(x \leq x_\alpha) = F(x_\alpha) = \alpha$$

Quantiles, Median and Mode



Selecting Mean, Median, and Mode

